UE26CY151A Engineering Chemistry · Unit 2 · Electrochemistry and Corrosion

Graphs and diagrams

Every graph and diagram in scope, redrawn as clean vector figures.

Class 002 · Nernst equation and its derivation · compiled 17 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
added for clarity	Standard context that is not in the class material
transcript only	Said in the lecture but not on any slide
video only	Seen only in a frame of the class video
not on current slides	From older material or past papers, not this year's slides
S002·4	Deck 002, slide 4 (rendered .pptx)
T001 00:21:42	Class 001 AV-summary transcript (covers classes 1 and 2), segment start
PDF p.13	Consolidated Unit 2 slide PDF (older syllabus), page 13
ESA Jan-25 s1 2a	Past paper: Jan 2025, set 1, question 2.a

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The deck has no diagrams; all six figures are added for clarity and drawn with standard textbook values (E°: Zn −0.76 V, Cu +0.34 V, Al −1.66 V). Okabe–Ito colours: blue for the first series, vermilion for the second, green for the third; grey dashed lines are references ($E = E^\circ$, $Q = 1$). Cell sketches are schematic, not to scale.

1 Free energy and electrode potential

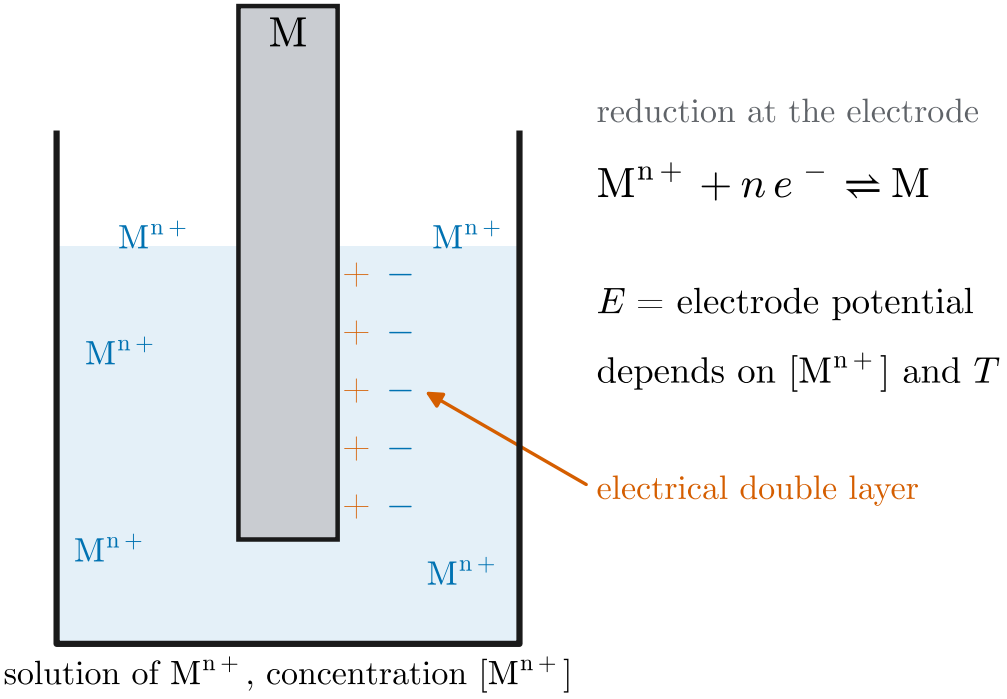


Figure G1.1. A single electrode: metal M in a solution of its ions. ESA ★★

ADDED FOR CLARITY S002·3 · T001 00:19:56

How to read it. The system of the derivation: a rod of metal M dipped in a solution of M^{n+}. The electrode reaction is written as a reduction. The charge separation at the interface (the electrical double layer from class 001) gives the electrode potential E, which the Nernst equation links to $[M^{n+}]$ and T. Schematic, not to scale.

3 Using the single-electrode equation: concentration and temperature

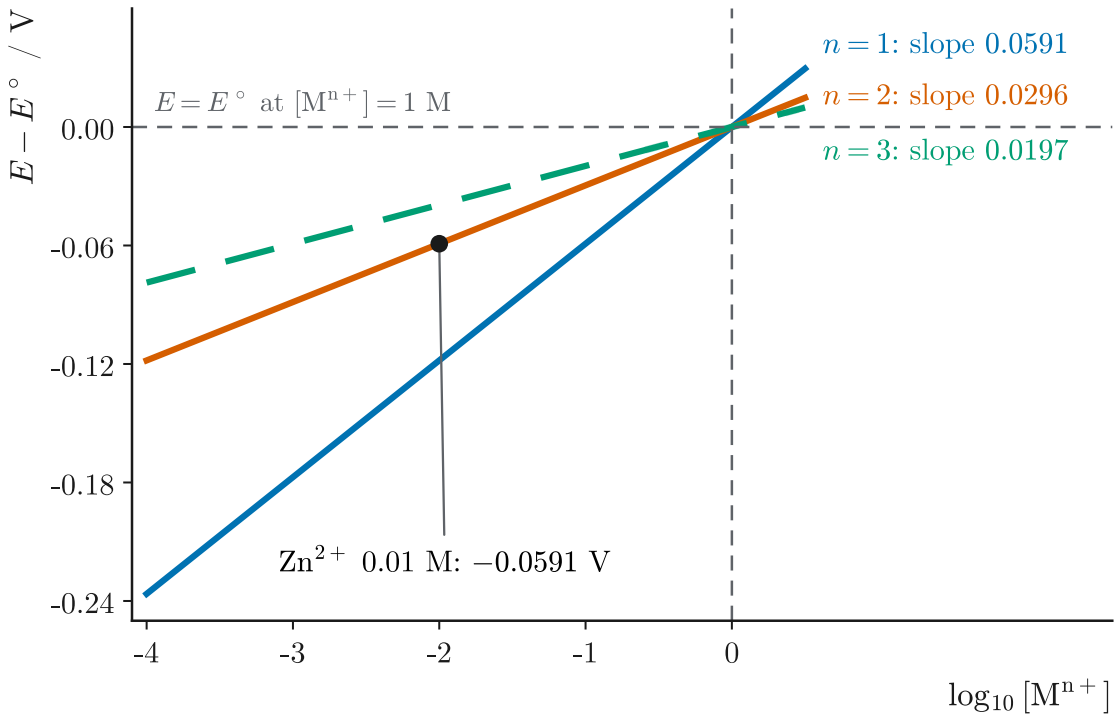


Figure G3.1. $E - E^\circ$ against $\log[M^{n+}]$ at 298 K. ESA ★★

ADDED FOR CLARITY S002·5

How to read it. Straight lines through the origin ($E = E^\circ$ at 1 M) with slope $0.0591/n$: 59.1 mV per decade for $n = 1$, 29.6 mV for $n = 2$, 19.7 mV for $n = 3$ (dashed). The marked point is Zn^{2+} at 0.01 M (two decades below 1 M, $n = 2$): -0.0591 V, as in W3.1.

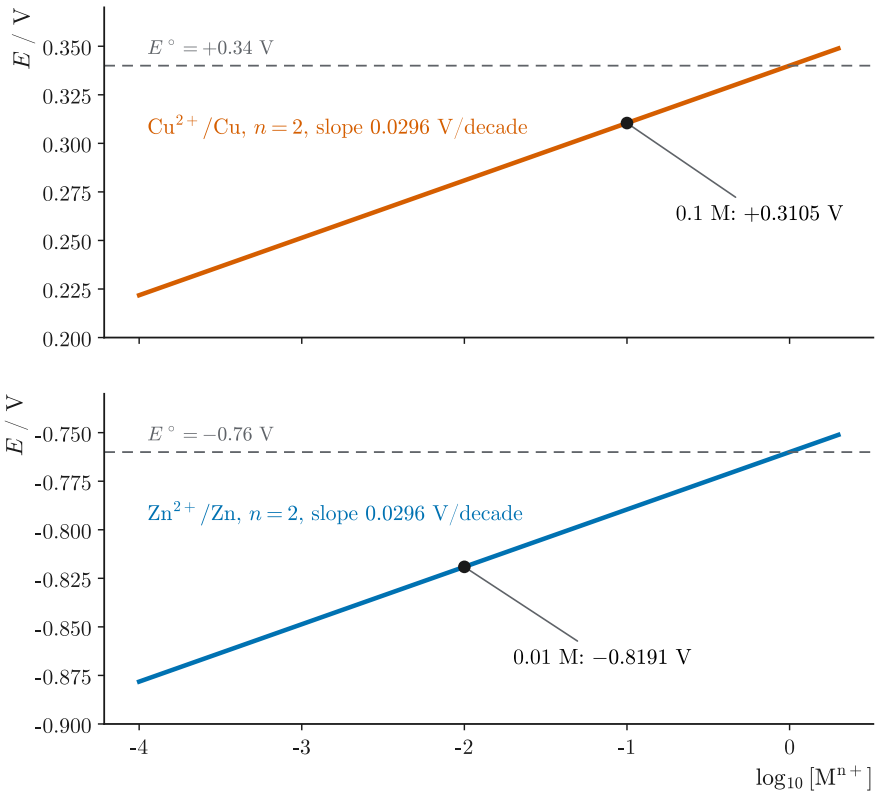


Figure G3.2. Electrode potentials of Cu and Zn against $\log[M^{n+}]$. ESA ★★

ADDED FOR CLARITY S002·5

How to read it. Each panel has its own vertical range (the lines are 1.10 V apart, which is E°_{cell} of the Daniell cell). Both have slope **0.0296** V per decade ($n = 2$). The grey dashed line is E°; the marked points are W3.2 (Cu, 0.1 M: +0.3105 V) and W3.1 (Zn, 0.01 M: −0.8191 V).

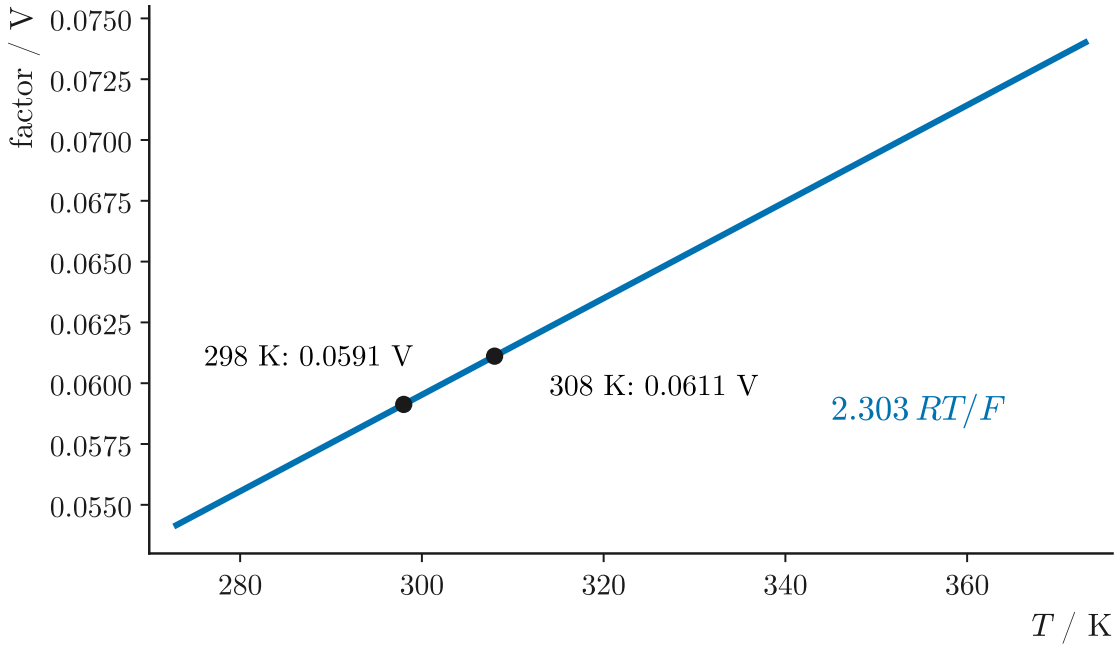


Figure G3.3. The Nernst factor $2.303RT/F$ against temperature. ESA ★

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How to read it. The factor is proportional to T (slide constants): 0.0591 V at 298 K and 0.0611 V at 308 K. Use the value for the question's temperature; dividing by n gives the slope of G3.1.

4 Nernst equation for a cell

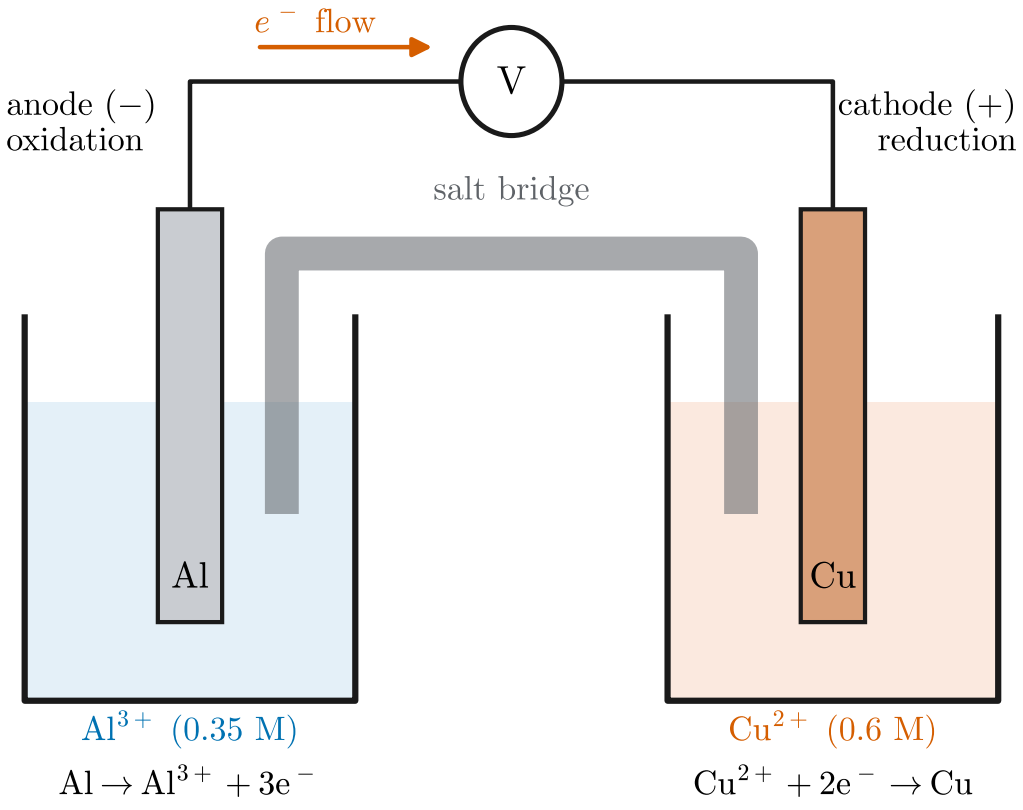


Figure G4.1. The Al–Cu cell of ESA Jan 2025. ESA ★★★

ADDED FOR CLARITY ESA Jan-25 s1 2a

How to read it. Al (lower E°, −1.66 V) is the anode and is oxidised; Cu (+0.34 V) is the cathode. Electrons flow through the wire from Al to Cu; the salt bridge completes the circuit. Cell: $\mathrm{Al}\,|\,\mathrm{Al^{3+}}(0.35\,\mathrm{M})\,\|\,\mathrm{Cu^{2+}}(0.6\,\mathrm{M})\,|\,\mathrm{Cu}$, $E_{cell} = 2.0024$ V. Schematic, not to scale.

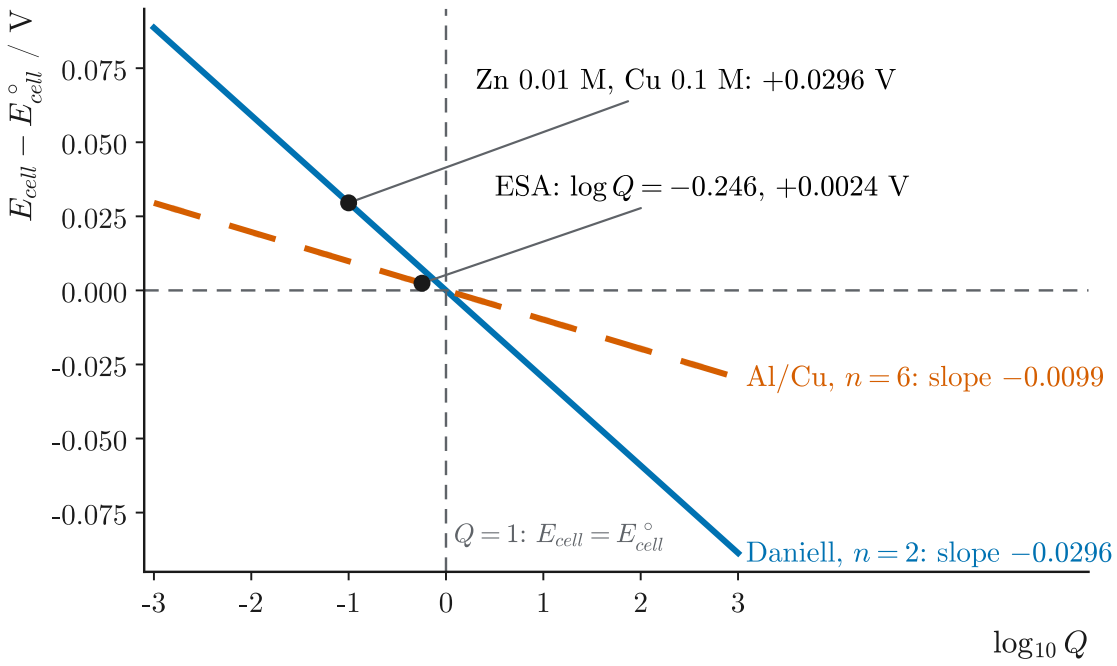


Figure G4.2. $E_{cell} - E^\circ_{cell}$ against $\log Q$. ESA ★★

ADDED FOR CLARITY S002·5 · ESA Jan-25 s1 2a

How to read it. Slope $-0.0591/n$: the larger n, the flatter the line, so the Al–Cu cell ($n = 6$, dashed) barely moves from 2.0 V. $Q < 1$ (left) raises the emf above E°_{cell}; $Q > 1$ lowers it. Points: W4.1 and W4.2.